~~++++~~

$f = [3, 2, 4, 6]$

dimensions of f → [4]

Shape of f

Δf = differences = Delta f

$= [2-3, 4-2, 6-4]$

$= [-1, 2, 2]$

Dot Product

$x + 2y = 3$

$2x - 4y = 5$

$\begin{pmatrix} 1 & 2 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$

$(1, 2) \cdot (x, y) = 3$

$(2, -4) \cdot (x, y) = 5$

$u \cdot v = \sum_{i=1}^{n} u_i v_i$

Matrix Product

$A \cdot B$

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$

rows → ← Columns

rows on left dot Columns on right

$1 \cdot 5 + 2 \cdot 7 \quad 1 \cdot 6 + 2 \cdot 8$

$3 \cdot 5 + 4 \cdot 7 \quad 3 \cdot 6 + 4 \cdot 8$

$C_{ij} = \text{row } i \cdot \text{col } j$

$= \sum_{k=1}^{n} a_{ik} b_{kj}$

Cols A = rows B

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

i is any index i is a row of A

i is a col of B

$$C = A \cdot B$$

$m \times n$ A has m rows, n columns

C has m rows and l cols

B is $n \times l$ — Same: n rows, l columns

$AB \neq BA$

X has 5 rows 3 columns

Y has 4 rows 3 columns

$X \cdot Y$ not allowed $4 \neq 3$

Y^T has 3 rows 4 colums

$X \cdot Y^T$ is legal

$Y \cdot X^t$ is legal

Transpose

$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

not reshaping

5 time steps (1, 2, 3 / 0, 1, 2) 3 numbers MD, VA, DE

row = year
col = stat

```
[ [ [10.1, 12.3, 1.5],   0
    [10.2, 12.4, 1.6],   1
    [10.3, 12.5, 1.7],   2
    [10.4, 12.6, 1.4],   3
    [10.3, 12.4, 1.5] ]  4
```

5 rows
3 cols
[5, 3]

A =
B =

B[3][2] = 1.4
0 to 4 0 to 2

A · B what dim's allowed for A?

M x 5 5 x 3

Cols of A = 5
m could be anything
A · B m rows and 3 cols

B^t = [[10.1, 10.2, 10.3, 10.4, 10.3], row = state, col = year
[12.3, 12.4, 12.5, 12.6, 12.4],
[1.5, 1.6, 1.7, 1.4, 1.5]]

$A \cdot B^t$
cols A = 3

2·MD + 3·VA − DE row of A (2, 3, −1) for each year

[12] [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]

[3,4] [[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12]]

3x2x2 [[[1,2], [3,4]], [[5,6], [7,8]], ...

[[[1, 2, 3, 4], [5, 6, 7, 8]], [[...

[illegible]